

ReCell AI — The Global Intelligence Platform for Second-Life Battery Markets

Date: May 3, 2026

Time: Morning Drop

Sector: Clean Energy / Circular Economy / Energy Security

Stage: Pre-Seed Ready

The \$50B Problem

The world is sitting on a ticking time bomb — and a goldmine.

300+ GWh of EV batteries will retire by 2030. These batteries aren't dead — they retain 70-80% capacity. They're perfect for stationary energy storage. But there's no standardized way to:

- **Grade** retired batteries for remaining capacity and health
- **Match** them to appropriate second-life applications
- **Trade** them efficiently in a transparent marketplace
- **Certify** their safety and performance for insurance/financing
- **Track** their provenance and remaining value over time

Meanwhile, geopolitical instability (the Iran situation) is driving unprecedented demand for **energy independence**. Businesses, homes, and communities want backup power and grid resilience. They can't afford new battery systems at \$400-600/kWh.

Second-life batteries at \$80-150/kWh could democratize energy storage. But the market is fragmented, opaque, and riddled with information asymmetry.

The Vision

ReCell AI is the intelligence and exchange layer for second-life batteries — the “Bloomberg Terminal + CARFAX + NYSE” for the battery aftermarket.

We create the trust infrastructure that enables a \$50B+ market to function:

1. **Universal Battery Intelligence** — AI-powered grading, health assessment, and remaining useful life prediction
 2. **Global Exchange** — Transparent marketplace connecting battery sellers (automakers, fleet operators, recyclers) with buyers (energy storage developers, businesses, utilities)
 3. **Certification & Insurance** — Standardized trust credentials that enable financing and insurance for second-life deployments
 4. **Provenance Tracking** — Blockchain-backed chain of custody from vehicle to second life to eventual recycling
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Why Now? (The Perfect Storm)

1. The Iran War & Energy Security Crisis

Geopolitical instability has made energy independence a national security priority. FT reports investors are “piling into clean energy” as the conflict drives the push for energy security. Home and business energy storage demand is exploding.

2. The Second-Life Battery Wave

The first mass wave of EV batteries is hitting retirement age. By 2030, retired EV batteries could provide more storage capacity than the entire current grid battery market.

3. The 7x Innovation Surge

European Patent Office reports battery reuse and recycling innovations increased 7x in the last decade. The technology for second-life applications is maturing rapidly.

4. The Price Gap Problem

New lithium-ion batteries: \$139/kWh (2025) → still too expensive for widespread deployment
Second-life batteries: \$80-150/kWh potential → game-changing economics
But only if buyers can trust what they're getting.

5. Regulatory Tailwinds

- EU Battery Passport regulations (2027) requiring lifecycle tracking
- US IRA incentives for domestic energy storage
- Emerging state-level mandates for battery circularity

Product Architecture

Layer 1: Universal Battery Intelligence Engine

The Problem: Every battery pack is a black box. Different chemistries, manufacturers, usage histories, and degradation patterns.

The Solution: AI-powered assessment that creates standardized “Battery Health Scores”

- **Non-Invasive Diagnostics:** Partner with battery management system (BMS) providers to access charge/discharge data without disassembly
- **Predictive Degradation Models:** ML models trained on millions of battery cycles to predict remaining useful life
- **Application Matching:** Algorithm that matches battery characteristics to optimal second-life use cases (frequency regulation vs. peak shaving vs. backup power)

Output: A “CARFAX Report” for every battery — standardized, verifiable, insurable.

Layer 2: The Global Exchange

Multi-Sided Marketplace:

Sellers	Buyers	Facilitators
Automakers (OEM take-backs)	Energy storage developers	Installers & integrators
Fleet operators (Hertz, Amazon)	Commercial & industrial facilities	Financiers & insurers
Dealerships & collision centers	Utilities & grid operators	Recyclers (end-of-life)
Recycling companies	Residential (via installers)	Logistics providers

Exchange Mechanics: - Real-time bid/ask pricing by battery grade, chemistry, and location - Auction and fixed-price sale options - Escrow and quality guarantee (ReCell Verified) - Integrated logistics coordination

Layer 3: Certification & Trust Infrastructure

ReCell Certified™ — The trust credential that unlocks financing and insurance:

- Standardized testing protocols (partnering with UL, TÜV, national labs)
- Performance warranties backed by actuarial data
- Insurance products (battery performance insurance, fire liability)
- Financing facilitation (equipment loans, leasing structures)

Layer 4: Provenance & Compliance

Battery Passport Integration: - Digital twin for every battery tracking full lifecycle - EU Battery Passport compliance built-in - Chain of custody from OEM → first life → second life → recycling - Carbon accounting (embedded carbon, avoided emissions)

Business Model

Revenue Streams

Stream	Model	Year 3 Target
Transaction Fees	3-5% of GMV on exchange	\$45M
Intelligence SaaS	Subscription for battery grading/assessment	\$25M
Certification Fees	Per-battery certification + annual verification	\$15M
Insurance Premiums	Commission on battery insurance policies	\$10M
Data Products	Market intelligence, pricing indices, analytics	\$8M
Financing Facilitation	Referral fees for equipment financing	\$7M

Total Year 3 Revenue Target: \$110M

Unit Economics

- **Average Transaction Size:** \$15,000 (battery pack or pallet)
- **Take Rate:** 4% blended
- **Customer Acquisition Cost:** \$500 (B2B sales motion)
- **Lifetime Value:** \$12,000 (repeat transactions + SaaS)
- **LTV:CAC:** 24:1

Go-to-Market Strategy

Phase 1: Supply Aggregation (Months 1-12)

“Control the supply, own the market”

- Partner with 3-5 major automakers for OEM take-back batteries

- Sign exclusive agreements with top 10 fleet operators
- Integrate with collision repair networks (Caliber, Gerber)
- Build the intelligence engine with real battery data

Target: 50,000 batteries in inventory/pipeline

Phase 2: Demand Activation (Months 6-18)

“Create the buyer network”

- Partner with commercial solar installers (Sunrun Commercial, Ameresco)
- Direct sales to Fortune 1000 facilities teams
- Utility partnerships for grid-scale procurement
- Launch installer certification program

Target: 500 active buyers, 10,000 batteries transacted

Phase 3: Market Dominance (Months 12-36)

“Become the index”

- Launch ReCell Price Index (the benchmark for second-life pricing)
- Expand to Europe (EU Battery Passport mandate)
- Insurance and financing products at scale
- Recycler integration (close the loop)

Target: \$1B GMV, 30% market share in tracked geographies

Competitive Landscape

Direct Competitors

Company	Approach	Gap
Li-Cycle	Recycling focus	Skips second-life value capture
Redwood Materials	Recycling/materials recovery	Manufacturing focus, not marketplace
Spiers New Technologies	Second-life refurbishment	Limited scale, no exchange
B2U Storage Solutions	Utility-scale deployment	Vertical, not horizontal platform

Our Moat: We’re not competing with recyclers or deployers — we’re the **intelligence and exchange layer** that makes the entire ecosystem more efficient.

Indirect Competitors

- OEM captive programs (limited to own batteries)
 - Brokers and traders (fragmented, no standardization)
 - Battery testing companies (point solution, no marketplace)
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Defensibility

Network Effects

- More sellers → better selection → more buyers → more sellers
- More transaction data → better ML models → better grading → more trust → more transactions

Data Moat

- Largest dataset of battery health and degradation across manufacturers
- Predictive models improve with every battery assessed
- Price indices become industry standard

Regulatory Positioning

- First mover on EU Battery Passport integration
- Compliance infrastructure that becomes mandatory

Switching Costs

- Seller inventory management integration
 - Buyer procurement workflow integration
 - Historical transaction data and analytics
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Team Requirements

Founding Team Profile

CEO: Enterprise marketplace experience (Flexport, Faire, or similar B2B marketplace) **CTO:** Battery diagnostics + ML background (Tesla BMS, battery analytics startup) **COO:** Commodities trading operations (ICE, CME, or commodity trading house) **Chief Science Officer:** Battery degradation research (Argonne, NREL, university lab)

Key Early Hires

- Head of OEM Partnerships (automotive industry relationships)
 - Head of Certification (UL, TÜV, or insurance industry background)
 - Lead ML Engineer (predictive modeling)
 - Head of Sales (B2B energy sales)
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Financial Projections

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
GMV	\$50M	\$250M	\$1B	\$3B	\$7B
Revenue	\$5M	\$25M	\$110M	\$280M	\$600M
Gross Margin	60%	65%	70%	72%	75%
EBITDA	-\$8M	-\$5M	\$15M	\$75M	\$200M
Headcount	25	75	200	400	700

Funding Strategy

Seed Round: \$8M

- Build core intelligence engine
- Sign first 3 OEM partnerships
- Launch beta marketplace
- Hire founding team (15 people)

Target Investors: Climate tech specialists (Congruent, Lowercarbon, Prelude)

Series A: \$35M (Month 18)

- Scale marketplace operations
- Expand to 10+ supply partners
- Launch certification program
- European market entry

Target Investors: Marketplace specialists + climate (a]6z, Ribbit, Union Square)

Series B: \$100M (Month 36)

- Market dominance push
- Insurance and financing products
- Full EU Battery Passport integration
- Asia expansion

Risk Analysis

Risk	Probability	Impact	Mitigation
OEMs build captive programs	Medium	High	Multi-OEM aggregation value; offer white-label option
Battery prices drop faster than expected	Medium	Medium	Second-life premium over recycling still exists
Safety incident from second-life battery	Low	Critical	Rigorous certification; insurance products
Regulatory changes	Low	Medium	Active policy engagement; compliance-first design
New battery tech makes reuse unnecessary	Low	High	Chemistry-agnostic platform; recycling integration

The Billion-Dollar Thesis

By 2035, the second-life battery market will be \$50B+ annually.

ReCell AI captures value at every transaction: - 100M+ retired EV battery packs globally - Average value: \$500-5,000 per pack - Multiple transactions per battery lifecycle (first sale, lease, resale) - Recurring certification and insurance revenue

If we become the “NYSE + S&P” for second-life batteries: - 5% of a \$50B market = \$2.5B GMV take - 4% take rate = \$100M+ annual revenue from transactions alone - Plus SaaS, data, insurance, financing = **\$500M+ revenue business** - At 15x revenue multiple = **\$7.5B valuation**

The Ask

We’re raising \$8M Seed to build the trust infrastructure for the second-life battery economy.

The timing is perfect: - Geopolitical crisis driving energy independence demand - First mass wave of EV batteries hitting retirement - 7x innovation surge in battery reuse technology - Regulatory tailwinds from EU Battery Passport

The window is now. The batteries are coming. We’re building the exchange.

Appendix: Market Size Calculation

Total Addressable Market (TAM)

- Global energy storage market: \$500B by 2035
- Second-life batteries addressable: 20% = \$100B

Serviceable Addressable Market (SAM)

- US + EU markets initially: \$50B
- Assessment, exchange, certification layers: \$15B

Serviceable Obtainable Market (SOM)

- Year 5 target: \$3B GMV (6% of SAM)
 - Revenue capture: \$300M (10% of GMV through all revenue streams)
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Contact

ReCell AI

The Intelligence Platform for Second-Life Battery Markets

Building the trust infrastructure for the \$50B battery aftermarket.

“The greenest battery is the one already made. We’re building the market to give it a second life.”